INCH-POUND

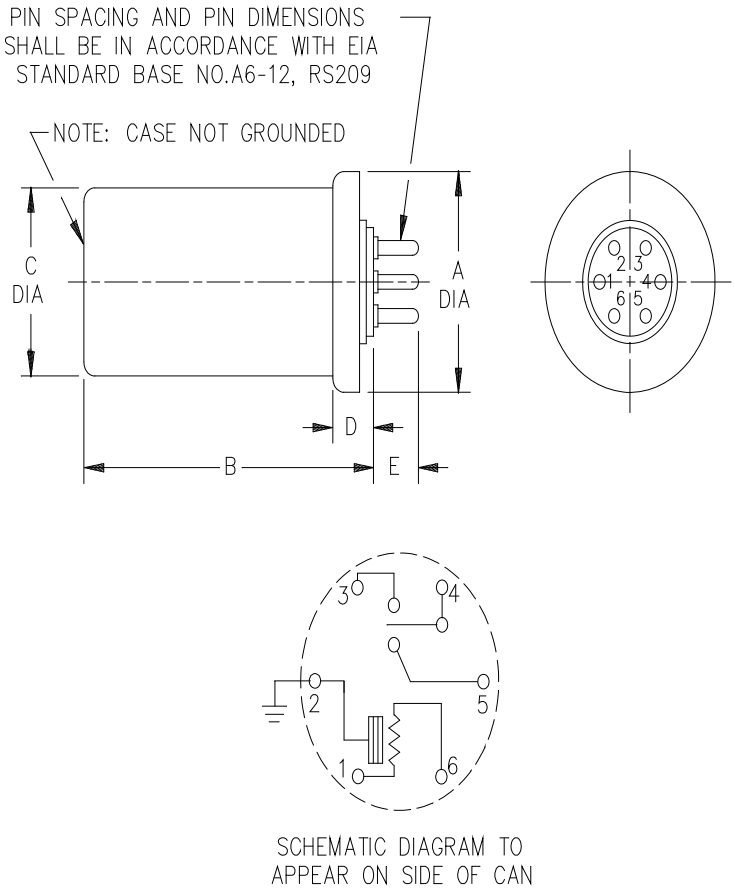
MIL-DTL-83729/2D
04 March 2013
SUPERSEDING
MIL-DTL-83729/2C
14 June 2002

DETAIL SPECIFICATION SHEET

CHOPPERS, ELECTROMECHANICAL, DUST TIGHT, SINGLE POLE, DOUBLE THROW, MAKE BEFORE BREAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83729.



Letter	Inches	mm
A	MAX 2.344	MAX 59.54
B	MAX 4.000	MAX 101.60
C	MAX 2.078	MAX 52.78
D	MAX .562	MAX 14.27
E	MAX .625	MAX 15.88

FIGURE 1. Outline drawing, terminal arrangement and schematic diagram.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (.13 mm).
4. When the coil is de-energized the center reed may be in any of 3 positions.

FIGURE 1. Outline drawing, terminal arrangement and schematic diagram - Continued.

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REQUIREMENTS:

Contact data:

Single pole, double throw, make before break.
Polarity: When pin 1 is positive, pin 4 is connected to pin 5 and disconnected from pin 3. When pin 6 is positive, pin 4 is connected to pin 3 and disconnected from pin 5.
Voltage and current: 1 microvolt to 6.0 volts, maximum. .001 ampere maximum

Coil data:

Voltage rated: 6.3 volts maximum.
Frequency range: 50 to 65 Hz.
Resistance: 131 ohms ± 10 percent.
Impedance: 132 ohms ± 10 percent.

Waveform characteristics (at 25°C):

Dwell time: Each contact closed 198° ± 7° of each cycle. Contacts closed simultaneously for 18° ± 10 percent twice in each cycle.
Dissymmetry: Within ± 7°.
Transfer time: Not applicable.
Phase angle: 17° ± 5°.

Noise level: Stray pick up.

Electrostatic: 2×10^{-10} V/ohm of input circuit impedance.
Electromagnetic: Less than 2×10^{-6} volts constant to within 2×10^{-7} volts.

Environmental conditions:

Dielectric withstanding voltage: Applicable.
Insulation resistance: Applicable.
Vibration II: Method 204, test condition A, MIL-STD-202.
(a) Qualification and quality conformance inspection: 1 1/2 hours in each of three mutually perpendicular directions, including the most critical axis.
Output voltage: Vary less than 2 percent from 0 to 10 g's during vibration.
Shock: 30 g's for 11 milliseconds in three directions.
Temperature range: 25°C ± 10°C.

Life:

Duration of test: 2,500 hours continuous.
Temperature: 25° ± 2°C.
Coil voltage and frequency: Nominal.
Operating conditions: See table I.
Output voltage: Vary not more than 2 percent during life.
Contact resistance: Shall not exceed .100 ohm and at conclusion of life.

TABLE I. Life operating conditions.

Contact voltage (volt)	Contact current (ampere)	Hours
.010	.001	1,000
1.0	.0001	1,500

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Mechanical:

Weight: .68 pound, maximum.
Dimensions: See figure 1.
Enclosure: Dust-tight case.
Terminals: Plug in, see figure 1.
Mounting position: Vertical for optimum performance.

Part or Identifying Number (PIN): M83729/2-001.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-83729, this document references the following:
MIL-STD-202

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5945-2013-012)

Review activities:
Army – AT, AV, CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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